Distance Decay as a Dominant Transferable Factor: Zero-Shot Gravity Model Calibration from Trip-Length Distributions

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Abstract

Predicting commuting flows without access to individual travel trajectories is a major challenge in urban planning. While deep learning methods achieve high accuracy, they require localized origin-destination (OD) surveys for calibration, limiting their utility in data-scarce or privacy-restricted domains.

This study investigates whether trip-length distributions (TLDs)—aggregate, population-level histograms of traveled distances—are empirically sufficient to recover city-specific distance-decay parameters for OD-free gravity model calibration. We evaluate two scenarios: an *OD-free calibration* setting utilizing target-city TLDs without exposing individual zone-to-zone records, and a *pure zero-shot fallback* employing a source-derived national decay prior.

Our framework is evaluated across 50 US metropolitan areas with 25/25 test. The parameters recovered from aggregate TLDs correlate strongly with the power-law exponent fitted from complete OD matrices and moderately with the exponential rate. Under oracle outflows, recovered parameters yield a negligible performance gap in downstream flow predictions.

Through a game-theoretic Shapley value analysis, we show that the distance-decay component dominates predictive capacity (accounting for 85.8% of the total CPC gain), suggesting that it acts as the dominant transferable factor in spatial interaction. The proposed model using target-city Meta Movement Distribution Maps (MDM) decay achieves downstream performance statistically equivalent to the locally-calibrated traditional Tanner baseline within a 0.01 CPC margin. In fully survey-free evaluations under predicted outflows, although the model's accuracy remains below high-capacity, black-box neural benchmarks, it provides a highly interpretable, data-efficient, and privacy-preserving alternative.

Keywords: Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

1 Introduction

Predicting origin-destination (OD) travel flows is essential for urban transport planning and infrastructure design (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). However, traditional travel surveys and proprietary cellular records are highly restricted due to cost, commercial licensing, and privacy concerns. Consequently, a central challenge in urban mobility modeling is the zero-shot cross-city transfer problem: predicting commuting flows in a target city where no local OD surveys are available.

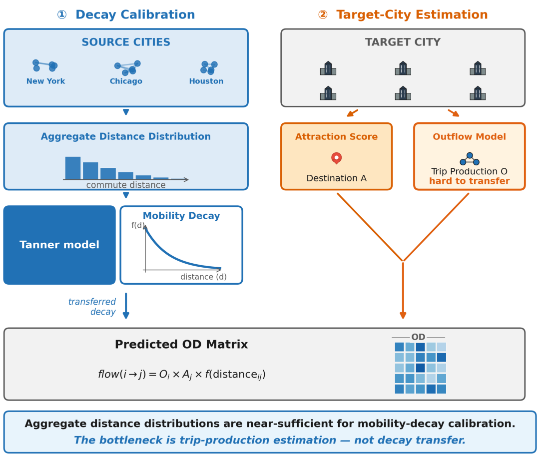

Fig. 1 Overview of the core findings: Aggregate trip-length distributions contain sufficient information to recover city-specific distance decay $f(d)$, which acts as a dominant transferable factor for predicting zero-shot flows within the investigated gravity framework. Prediction performance is bounded by the model's structural gravity assumptions.

While recent deep learning approaches achieve high accuracy, they require abundant individual-level OD records for training, limiting their generalizability and raising privacy concerns. In contrast, classical gravity models (Wilson 1971) offer physical interpretability and structural simplicity but require local calibration of their distance-decay functions. This raises a fundamental scientific question: **Can aggregate mobility distributions, which do not reveal individual OD pairs, serve as a reliable calibration signal for recovering city-specific distance-decay functions?**

In this paper, we address this question by proposing a three-tier conceptual framework. First, we identify the city-specific distance-decay function as the *dominant transferable factor* of spatial interaction. Second, we utilize the target-city aggregate trip-length distribution (TLD) as an *aggregate representation that reduces privacy exposure*, allowing this decay function to be identified without exposing individual zone-to-zone trajectories. Third, we apply this recovered decay in a *zero-shot calibration* setting to predict commuting flows. Under the primary **OD-free calibration setting**, the model recovers distance-decay parameters from target-city aggregate TLDs (e.g., Meta Movement Distribution Maps). In the **pure zero-shot fallback**, the model uses a national decay prior derived from source cities. We show that aggregate trip-length histograms contain sufficient information to recover the city-specific distance-decay structure with high fidelity. Furthermore, our game-theoretic Shapley analysis attributes the vast majority of predictive capacity to distance decay within the investigated framework, which is consistent with the interpretation that distance decay functions as the dominant transferable factor of spatial interaction.

To investigate these relationships systematically, we organize the investigation around three research questions:

1. Can aggregate trip-length distributions recover the city-specific distance-decay structure without access to individual OD trajectories?
2. How does distance decay impact zero-shot gravity transfer of commuting flows, and to what extent does its recovery from trip-length distributions drive transfer performance?
3. Under what urban morphological conditions does aggregate-calibrated decay perform most effectively, and what physical geography factors govern those conditions?

Our empirical investigation across 50 US metropolitan areas evaluates these assumptions. The results indicate that distance decay represents a dominant transferable factor across metropolitan areas, and that projecting spatial interactions onto the TLD provides a summary representation sufficient to identify this component. Although the proposed model's overall predictive accuracy remains below the black-box neural DeepGravity benchmark (average CPC 0.6860 vs. 0.7529), it achieves strong performance across diverse urban morphologies, while remaining completely survey-free, highly interpretable, and privacy-compliant. The rest of this paper is organized around the three research questions detailed above.

1.1 Classical and Structured Mobility Models

Classical spatial interaction models, such as gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012), form the theoretical foundation of urban flow modeling. Gravity formulations relate traffic flows directly to zone masses and inversely to travel distance, while radiation models simulate destination choice based

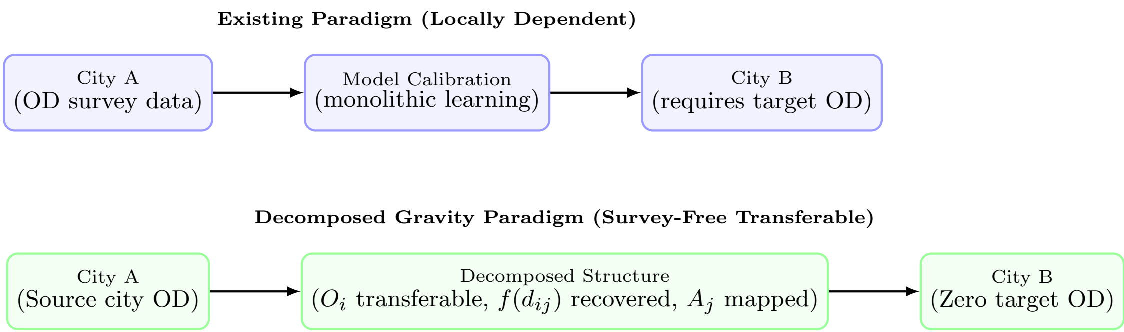

Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus the decomposed gravity paradigm investigated in this study. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In the decomposed paradigm, transferable origin production structures are learned from source cities, decay is recovered from target aggregate data, and attraction is mapped from local open features, enabling survey-free transfer without target-city OD surveys.

on the distribution of intervening opportunities. Although highly interpretable due to explicit physical constraints, gravity models require local origin-destination (OD) surveys to calibrate their distance-decay parameters. Meanwhile, radiation models tend to perform poorly in urban regions characterized by highly directional commuting patterns or constrained transport networks.

1.2 Deep Learning for OD Prediction

To address the limitations of classical frameworks, deep learning techniques have been introduced to capture non-linear relationships in spatial interactions. Models like DeepGravity (Simini et al. 2021) employ deep neural networks to map built-environment variables directly to flow volumes, and subsequent works incorporate graph neural networks (Rong et al. 2023) or OpenStreetMap representations (Atwal et al. 2025). Although these high-capacity neural models achieve superior predictive performance when training data is abundant, they require extensive individual-level OD records for local adaptation, which restricts their generalizability to data-poor or privacy-sensitive cities.

1.3 Aggregate Mobility Data for Decay Calibration

To establish data-efficient modeling pipelines, researchers have explored aggregated mobility datasets that minimize privacy risks. These datasets can be compressed into trip-length distributions (TLDs), which represent one-dimensional marginal histograms of travel distances without disclosing individual zone-to-zone trajectories (Pappalardo et al. 2023). Although TLD shapes correspond to standard parametric decay functions (Brockmann et al. 2006; Lenormand et al. 2016), the literature has not yet addressed how to leverage these aggregate distributions as a calibration signal to guide zero-shot cross-city transfer.

1.4 Cross-City Transfer Learning

Cross-city transfer learning applies predictive knowledge from data-rich source cities to data-scarce target cities. Current transfer methodologies rely on region representation alignment (Wang et al. 2018), generative geographic pretraining (Mendieta et al. 2023), or domain adaptation (Wang et al. 2022). Nonetheless, these methods generally assume that some target-city flow samples are accessible during training, or they fail to account for city-specific variations in spatial friction, limiting their effectiveness in zero-shot settings.

1.5 Research Gap and Positioning

Table 1 compares how existing model classes address the core challenge of zero-shot cross-city transfer. The comparison reveals three critical literature gaps: the lack of an aggregate-based decay calibration method for zero-shot transfer, the absence of a quantitative analysis regarding the impact of distance decay on zero-shot transfer, and unidentified urban morphological conditions that influence the performance of aggregate calibration. These gaps map directly to the research questions evaluated throughout this study.

2 Problem Setting

We formalize the task of zero-shot cross-city travel flow prediction as follows. Let a target metropolitan area be partitioned into a set of N discrete spatial zones (e.g., census tracts) denoted by $\mathcal{V} = \{1, \ldots, N\}$. Our objective is to predict the origin-destination (OD) flow matrix $\mathbf{T} \in \mathbb{R}_{+}^{N \times N}$, where each element T_{ij} represents the volume of seasonal commuting trips from an origin zone $i \in \mathcal{V}$ to a destination zone $j \in \mathcal{V}$.

In a traditional local calibration setting, models are optimized using detailed individual-level trajectory records or comprehensive travel surveys that represent the target city's true flow matrix $\mathbf{T}$. However, in the zero-shot cross-city transfer paradigm, we assume that no local zone-to-zone flow observations are available for the target metropolitan area. The model must instead predict the target-city flows using only globally available built-environment features, gridded demographic data, and varying levels of aggregate target mobility signals.

To analyze the performance of urban mobility prediction under different data constraints and privacy regulations, we define three operational and diagnostic evaluation settings, which are summarized in Table 2. These settings represent distinct scenarios of data availability:

- **Pure Zero-Shot Fallback**: This configuration assumes that absolutely no target-city mobility data is accessible. The model relies entirely on open spatial features and a national distance-decay prior derived from source cities.
- **OD-Free Calibration**: This is our primary proposed pipeline. The model has access to an aggregate, one-dimensional trip-length distribution (TLD) of the target city but cannot observe any individual origin-destination pairs.
- **Oracle Diagnostic Control**: This setting serves as an analytical benchmark to isolate the errors of the distance-decay recovery from those of the origin production predictions. In this control, the target-city origin outflows are set to their ground-truth values.

3 Methodological Design

The methodological design of this study is grounded in a separable formulation of the classical spatial gravity framework. Under this formulation, the commuter flow between an origin zone and a destination zone is modeled as a multiplicative product of distinct spatial components. This decomposition allows us to isolate the dominant transferable factors of human mobility from local, city-specific variables:

$$T_{ij} = O_i \cdot \frac{A_j f(d_{ij})}{\sum_k A_k f(d_{ik})} \tag{1}$$

where flows are decomposed into origin production (O_i), destination attraction (A_j), and distance decay ($f(d_{ij})$). We structure our investigation around three core hypotheses: (1) distance decay represents the dominant transferable factor of spatial interaction across cities; (2) aggregate trip-length distributions (TLDs) contain sufficient information to recover this decay structure without requiring individual origin-destination records; and (3) the recovered decay parameters can be integrated with machine learning models for production and open-data heuristics for attraction to enable survey-free target-city flow prediction.

3.1 Aggregate-Based Decay Calibration

We recover the distance-decay parameters $\theta = (\gamma, \beta)$ from the aggregate trip-length distribution (TLD) $p(d)$ — a marginal histogram of trip distances. Formally:

$$p(d) = \sum_{i,j} T_{ij} \cdot \mathbf{1}[d_{ij} \approx d] \tag{2}$$

Because the trip-length distribution is conditioned by zone geometries and employment opportunity arrangements, we formalize the parameter recovery using an exposure-corrected likelihood model. Let y_b be the observed commuter count in distance bin $b \in \{1, \ldots, K\}$ from the target-city TLD. Under the gravity framework, the predicted flow is:

$$\hat{T}_{ij}(\theta) = O_i \frac{A_j f_\theta(d_{ij})}{\sum_k A_k f_\theta(d_{ik})} \tag{3}$$

Table 1 Comparative analysis of human mobility models across structural and transferability dimensions. "Calibration-Free Decay" indicates whether the model recovers the distance decay function without requiring individual OD pair observations from target cities.

Model Class	Decomposed Structure	Cross-City Transfer	Zero-Shot Target	Interpretability	Calibration-Free Decay
Gravity (Wilson 1971)	✓	×	×	✓	×
Radiation (Simini et al. 2012)	×	✓	✓	✓	✓
DeepGravity (Simini et al. 2021)	×	Limited	×	×	×
GNN OD (Rong et al. 2023)	×	Limited	×	×	×
RegionTrans (Wang et al. 2018)	Partial	✓	×	×	×
SDF-ZTF (Ours)	✓	✓	✓	✓	✓

Table 2 Operational and diagnostic evaluation settings.

Setting	Target City Mobility Data	Deployment / Analytical Purpose
Pure Zero-Shot Fallback	None (Open spatial features only)	Fallback scenario under extreme data scarcity; uses national decay prior.
OD-Free Calibration	Aggregate TLD (no individual OD flows)	Primary proposed pipeline; enables local calibration while protecting privacy.
Oracle Diagnostic Control	Target outflows O_i controlled	Diagnostic benchmark; isolates decay recovery from production prediction errors.

where O_i represents the target outflow, A_j is the attraction heuristic (Section 3.4), and f_θ is the decay function. The predicted probability of a trip falling into bin b is:

$$P_\theta(b) = \frac{\sum_{i,j:d_{ij}\in b} \hat{T}_{ij}(\theta)}{\sum_{i,j} \hat{T}_{ij}(\theta)} \tag{4}$$

This predicted probability $P_\theta(b)$ explicitly corrects for the spatial opportunity distribution and zone geometries of the target city. Assuming a multinomial distribution of trips across the distance bins, we estimate $\theta = (\gamma, \beta)$ by maximizing the multinomial log-likelihood between the observed fractions $\bar{y}_b = y_b / \sum_{b'} y_{b'}$ and predicted probabilities $P_\theta(b)$:

$$\mathcal{L}(\theta) = \sum_{b=1}^{K} \bar{y}_b \log P_\theta(b) \tag{5}$$

Minimizing $-\mathcal{L}(\theta)$ recovers the parameters from aggregate data without individual zone-to-zone records. In production, O_i in Eq. (3) is replaced by the GBDT-predicted $\hat{O}_i$; the model structure is shown in Fig. 3.

3.2 Distance Decay Function

We model spatial friction using the Tanner multiplicative deterrence function (Tanner 1961), which combines power-law and exponential terms:

$$f(d_{ij};\, \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta\, d_{ij}) \tag{6}$$

where $\beta > 0$ represents the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ serves as an adaptive geometric offset.

The offset δ prevents singularity at $d_{ij} = 0$ and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N}\sum_{i=1}^{N} \sqrt{\frac{\text{Area}_i}{\pi}} \tag{7}$$

where $\bar{R}_{\text{zone}}$ is the mean zone radius. The parameters (β, γ) are estimated via MLE on the aggregate distance-bin distribution, with further optimization details provided in Appendix A. Because only binned trip-length histograms are used for calibration, this procedure is compatible with strict data-governance regulations (e.g., GDPR).

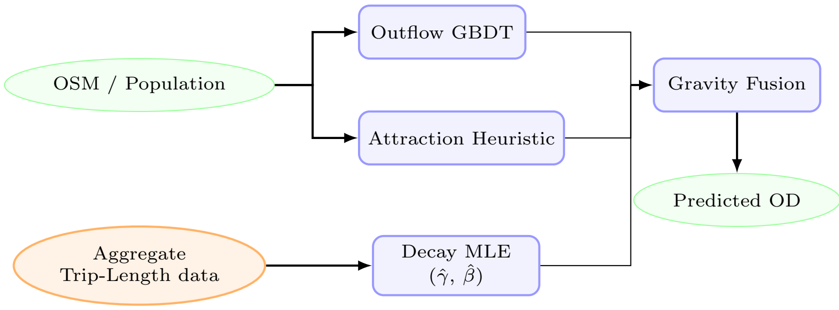

Fig. 3 Overview of the aggregate-calibrated gravity model structure. Globally available open-data features (green ellipses) are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) via a training-free min-max heuristic. The **Aggregate Trip-Length Histogram** (orange ellipse, dashed arrow)—which provides aggregate summaries that significantly reduce privacy exposure compared with individual trajectories—feeds the Decay MLE block to recover city-specific parameters $(\hat{\gamma}, \hat{\beta})$ without requiring any individual OD trajectories. The three components are combined via multiplicative gravity fusion to predict survey-free flow matrices (T_{ij}).

For numerical implementation and computational efficiency, the theoretical offset δ is regularized by clamping the minimum distance to a small epsilon ($d_{\min} = 10^{-6}$), which yields mathematically equivalent results as $d_{ij} \gg d_{\min}$ for all cross-zone flows ($i \neq j$).

3.3 Origin Outflow Component (O_i)

The trip production $O_i = \sum_j T_{ij}$ represents the total commuter outflow originating from zone i. In zero-shot transfer settings where target outflows are unobserved, we predict O_i using a Gradient Boosted Decision Tree (GBDT) regressor. The GBDT model is trained on a parsimonious set of **6 core macroscopic features**: total population (P_i), total POI count (POI_i), zone area (Area_i), population density ($\rho_{pop,i}$), POI density ($\rho_{poi,i}$), and road density (rd_i).

These 6 features were selected from an initial 37-candidate set using a cross-city SHAP stability analysis, showing no performance loss compared to the full feature set. This parsimonious feature set is chosen based on dimensional scaling in urban physics and Occam's razor, and avoids overfitting to source-domain geometries. Hyperparameters and details of the SHAP-based feature selection are provided in Appendix C.

3.4 Destination Attraction Component (A_j)

The destination attraction A_j represents a zone's capacity to draw trips. To prevent the model from overfitting to source-domain geometries in zero-shot settings, we define a training-free, scale-invariant attraction heuristic:

$$\hat{A}_j = \frac{1}{2}\left[\mathrm{minmax}(P_j) + \mathrm{minmax}(\mathrm{POI}_j)\right] \quad (8)$$

where $\mathrm{minmax}(x_j) = (x_j - \min_k x_k)/(\max_k x_k - \min_k x_k + 10^{-9})$, P_j is the gridded population count, and POI_j is the OpenStreetMap POI count. This formulation normalizes absolute scale differences across cities while preserving local opportunity hierarchies. In Section 4.2, the ablation analysis (Table 6) validates that this training-free heuristic contributes.

3.5 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

3.6 Data Sources

We utilize three globally available open data sources to compile features for our models: built-environment spatial descriptors (OpenStreetMap POIs and road density), gridded demography (WorldPop), and aggregate movement telemetry maps (Meta Movement Distribution Maps (Meta 2026)). Ground-truth origin-destination (OD) flows between census tracts are obtained from the LODES commuting dataset covering 50 US metropolitan areas (25 source cities for GBDT production training and 25 held-out target cities for testing). The target OD matrices are kept completely blind to the model components and are used exclusively as ground truth for validation.

Algorithm 1: Aggregate Calibration and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate trip-length histogram $\{y_b\}_{b=1}^{K}$ (or national prior if unavailable)
Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Offline Training (Source Cities)
1. **Production Model Training**: Train the outflow model $g(\mathbf{x})$ (GBDT) on pooled source-city spatial features $\mathbf{X}^{(c)}$ and ground-truth outflows $O_i^{(c)}$.

Phase 2: Online Calibration & Zero-Shot Transfer (Target City)
2. **Decay Calibration**: Recover target-city decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on the target city's aggregate distance-bin histogram $\{y_b\}_{b=1}^{K}$. (If unavailable, fall back to the source-derived national prior $(\beta_{\text{nat}}, \gamma_{\text{nat}})$).
3. **Attraction Mapping**: Compute target city zone attractions using the training-free heuristic: $\hat{A}_j^{(t)} = \frac{1}{2}\left(\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})\right)$.
4. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   c. Fuse components and normalize outflows: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)} \frac{\hat{A}_j^{(t)}(d_{ij}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}}}{\sum_k \hat{A}_k^{(t)}(d_{ik}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ik}}}$.

3.7 Meta MDM Harmonization

Meta MDM provides aggregate travel fractions at the county level grouped into 3 physical distance bins: $[0, 10$ km$]$, $(10, 100$ km$]$, and > 100 km. To harmonize these county-level travel fractions into tract geometries for calibration:

1. County-level trip-length distributions are averaged using WorldPop-derived population shares as weights: $p_{\text{metro}}(b) = \sum_{c \in \mathcal{C}} w_c \cdot p_c(b)$, where w_c is the population share of intersecting county c.
2. Tract pairs (i, j) are mapped to these 3 bins based on their centroid distances d_{ij}.
3. The predicted probability of a flow falling into Meta bin b is computed by summing and normalizing predicted tract-level flows in that bin: $P_\theta(b) = \sum_{i,j:d_{ij} \in \text{Range}(b)} \hat{T}_{ij}(\theta) / \sum_{i,j} \hat{T}_{ij}(\theta)$. This normalization bridges the continuous gravity formulation with Meta's 3-bin telemetry.

3.8 Urban Morphology Grouping

To investigate how physical geography modulates the role of distance decay in spatial interactions, the 25 target cities are classified into four morphological groups based on their urban structure and geography (González et al. 2008). This classification is summarized in Table 3.

4 Results

4.1 Can aggregate distributions recover distance decay?

The evaluation of downstream origin-destination (OD) flow prediction under the oracle outflow control demonstrates that employing the distance-decay parameters recovered from aggregate trip-length distributions (TLDs) yields predictive accuracy that closely matches the accuracy obtained with parameters fitted on full zone-to-zone OD surveys (Table 4). Specifically, the average difference in the Common Part of Commuters (CPC) index between using recovered parameters and using survey-calibrated parameters is 0.11%, with GBDT production prediction remaining the primary performance bottleneck in full zero-shot deployments (Table 7).

The parameter recovery performance varies between the two decay exponents, showing a correlation of $r = 0.893$ for the power-law exponent γ and $r = 0.582$ for the exponential rate β (Table 4).

4.1.1 Validation via Real-World Meta Mobility Prior

To verify the practical feasibility of using tech-platform telemetry, we evaluate the decay parameters recovered from Meta's actual Movement Distribution Maps prior (`Meta MDM Prior (3-Bin)`)

Table 3 Morphological classification of target cities.

Morphology Group	Cities	Description
Dense / Transit ($n = 5$)	Boston, Long Beach, New York, Philadelphia, Washington DC	High-density urban cores; extensive transit networks; weaker distance decay influence.
Sprawling / Grid ($n = 10$)	Atlanta, Dallas, El Paso, Houston, Jacksonville, Las Vegas, Mesa, Omaha, San Antonio, Tulsa	Low-to-medium density; regular road grids; car-dependent mobility with uniform decay.
Polycentric ($n = 7$)	Albuquerque, Charlotte, Detroit, Louisville, Milwaukee, Raleigh, San Jose	Multiple employment centers; flows cluster around local nodes rather than a single center.
Coastal ($n = 3$)	Portland, Seattle, Virginia Beach	Coastline/topographic barriers constrain the road network, amplifying distance friction.

Table 4 Decay parameter recovery performance metrics and downstream CPC under oracle outflow across the 25 target cities.

Parameter / Strategy	MAE	RMSE	Pearson r	Pearson r^2	Downstream CPC
Power-law exponent (γ)	0.0394	0.0549	0.893	0.798	–
Exponential decay rate (β, km^{-1})	0.0024	0.0052	0.582	0.338	–
Ground Truth (GT) Decay	–	–	Reference	–	0.7411
Recovered (Proposed) Decay	–	–	–	–	0.7403 (Gap: -0.11%)

across the 25 target cities. We compare its downstream CPC performance under oracle outflows against models calibrated on LODES commuting data aggregated into 3 bins (`LODES Baseline (3-Bin)`) and 20 equal-width bins (`LODES Oracle (20-Bin)`).

As shown in Table 5, the survey-free `Meta MDM Prior (3-Bin)` calibration yields a mean CPC of 0.7080 across the 25 target cities, compared to 0.7348 for the 3-bin survey baseline and 0.7403 for the 20-bin survey oracle.

Histogram compression is not the primary source of performance loss. When the original LODES commuting data is compressed from 20 bins to the same three physical distance bins used by Meta MDM (0–10 km, 10–100 km, > 100 km), CPC decreases only from 0.7403 to 0.7348 (-0.0055). This indicates that the coarse 3-bin representation preserves most of the information required for distance-decay recovery. In contrast, replacing the survey-derived histogram with real Meta MDM telemetry further reduces CPC from 0.7348 to 0.7080 (-0.0268). Therefore, most of the observed degradation is attributable to source mismatch and telemetry aggregation effects rather than to the limited number of bins. These results support the central claim that aggregate trip-length distributions remain highly informative for distance-decay calibration even under extremely compressed representations.

To evaluate performance under extreme data scarcity where target TLDs are completely missing, we evaluate the National Decay Fallback, where the national prior parameters are derived as the median of estimates fitted on source cities. Under oracle outflows, the National prior model achieves a CPC of 0.7398, and under GBDT-predicted outflows it yields a CPC of 0.6896 (Table 7).

The performance of the localized Meta MDM calibration is sensitive to local outlier dynamics. Specifically, Atlanta serves as an outlier, exhibiting a localized drop compared to the 3-bin baseline (Appendix A.2).

4.1.2 Decay Bin Sensitivity Analysis

Sweeping the distance bins $K \in \{3, 5, 10, 20\}$ shows that downstream CPC stabilizes at $K \geq 5$,

Table 5 Downstream CPC performance on the 25 held-out target cities under different decay calibration strategies using real Meta prior data.

Calibration Strategy	Data Source	Bin Configuration	Downstream CPC (Mean)
`LODES Oracle (20-Bin)` (Survey-Based)	LODES Commuting Data	20 Equal-Width Bins	0.7403
`LODES Baseline (3-Bin)` (Survey-Based)	LODES Commuting Data	3 Physical Bins	0.7348
`Meta MDM Prior (3-Bin)` (Survey-Free Prior)	Meta Movement Maps	3 Physical Bins	0.7080

with a CPC of 0.7174 for $K = 3$, 0.7346 for $K = 5$, 0.7394 for $K = 10$, and 0.7403 for $K = 20$ (Appendix A.2, Fig. 4).

4.2 Why is recovery sufficient?

To address research question 2, we first evaluate the origin production component (O_i) in isolation to validate the predictive quality of GBDT outflows, and then implement a game-theoretic Shapley value analysis to quantify the relative contributions of the model components.

4.2.1 Origin Production Transferability

To isolate the contribution of O_i, we hold A_j (OSM heuristic) and (γ, β) (national prior) fixed, then compare three outflow strategies across 25 target cities: a naive population baseline ($O_i \propto \text{Population}_i$), GBDT-predicted outflows, and oracle outflows.

We find that GBDT outflows (CPC 0.6896) beat the naive population baseline (0.6758) and achieve 93.2% of the oracle upper bound (CPC 0.7398) under zero-shot transfer. Robustness experiments are in Appendix B (Fig. 5).

Feature Selection via SHAP Stability Analysis. Evaluating GBDT model performance across nested subsets of top SHAP features indicates that zero-shot prediction performance plateaus at $K = 6$ features. The detailed feature taxonomy, SHAP stability criteria, and feature count sensitivity are provided in Appendix C (Fig. 6 and Fig. 7).

4.2.2 Factorial Ablation and Shapley Value Analysis

To evaluate the relative structural importance of each gravity component for zero-shot transfer, we conduct a factorial ablation across the 25 target cities. Each component is disabled in turn—replaced by a uniform-weight fallback—to isolate its marginal contribution under our gravity equations.

We implement a game-theoretic analysis using Shapley values (Shapley 1953) across all $2^3 = 8$ component coalitions. Over the uniform flow baseline (CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2258 (85.8% of total gain) to distance decay $f(d)$, 0.0208 (7.9% of total gain) to attractiveness A_j, and 0.0167 (6.4% of total gain) to outflow production O_i. Note that the factorial ablation baseline in Table 6 is the full model using the national decay prior (CPC 0.6896), where $f(d)$ removal yields a CPC of 0.4489.

4.3 Does aggregate-calibrated decay provide a practically deployable alternative to survey-based calibration?

To address the final research question, we evaluate whether our proposed aggregate-calibrated gravity model can serve as a deployable, survey-free alternative to traditional locally-calibrated frameworks. We compare our model against zero-shot neural architectures and locally calibrated structures across the 25 held-out target cities. The results of these comparative evaluations are summarized in Table 7.

An analysis of the multi-baseline comparison yields several key insights. First, we observe that predicting true origin outflows (O_i) does not significantly alter the predictive accuracy of the DeepGravity model (CPC 0.7526 vs. 0.7529), as its neural architecture jointly optimizes production and destination selection. In contrast, the proposed gravity model demonstrates a strong dependency on production accuracy, with the

Table 6 Factorial Ablation and Shapley Value Contributions (Average across $N = 25$ held-out target cities). The full-model baseline (top row) uses a fixed national decay prior ‡ to isolate component contributions.

$\mathbf{O}_i$	f(d)	$\mathbf{A}_j$	CPC	Factorial Contribution	Shapley Value
✓	✓‡	✓	0.6896	Baseline (all components, national prior decay)	–
×	✓‡	✓	0.6645	O_i removal: 3.6% loss	0.0167 (6.4%)
✓	×	✓	0.4489	f(d) removal: 34.9% loss	0.2258 (85.8%)
✓	✓‡	×	0.6574	A_j removal: 4.7% loss	0.0208 (7.9%)

‡In all ablation rows, $f(d)$ is frozen at the national prior $(\gamma_{\text{nat}}, \beta_{\text{nat}})$ to isolate the contribution of O_i and A_j. The full pipeline uses aggregate-recovered per-city decay (see Table 7).

Table 7 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Type	Features
Group 1: Survey-Free Deployment (No target OD data)				
Proposed Model (Survey-Free, Meta MDM)	0.6860	0.0391	Decomposed	6
Proposed Model (Survey-Free, National Fallback)	0.6896	0.0384	Decomposed	6
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	27
Group 2: Oracle Diagnostic (True outflows O_i^{true})				
Proposed Model (Oracle O_i, Meta MDM)	0.7337	0.0256	Decomposed	6
Proposed Model (Oracle O_i, National Prior)	0.7398	0.0248	Decomposed	6
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	27
Group 3: Locally-Calibrated Baselines (Fitted on target cities)				
Naive Population Baseline	0.6758	0.0384	Baseline	0
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	0
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	0
Radiation Model	0.4851	0.0516	Param-free	0

introduction of oracle outflows narrowing the performance gap and matching the locally-fit baseline.

To statistically validate these findings, we perform a Wilcoxon signed-rank test across the 25 target cities. The difference in CPC between our proposed model using Meta MDM decay (CPC 0.7337) and the locally-calibrated Traditional Tanner baseline (CPC 0.7382) is statistically significant but practically minor (raw $p = 0.00737$). Furthermore, a Two One-Sided Test (TOST) of equivalence confirms that our aggregate-calibrated model is equivalent to the locally-fitted Tanner baseline within a tight margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$). Additionally, the proposed model statistically outperforms the parameter-free Radiation Model baseline (Wilcoxon test with Bonferroni adjustment, $p < 10^{-6}$).

4.3.1 Exploratory Urban Morphology Analysis

To examine how different physical environments modulate model performance, we evaluate the proposed framework across the four morphological groupings described in Section 3.8.

Our analysis reveals that the physics-constrained gravity formulation achieves its highest predictive performance in Coastal environments (mean CPC of 0.714). The relative

Table 8 Survey-Free Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	Proposed CPC	DeepGravity CPC	Rel. Gap (%)
Dense / Transit	5	0.651	0.726	10.3%
Sprawling / Grid	10	0.691	0.760	9.0%
Polycentric	7	0.704	0.759	7.2%
Coastal	3	0.714	0.758	5.8%

performance gap between the proposed gravity model and the neural DeepGravity baseline narrows to 5.8% in Coastal cities, compared to a larger 10.3% gap in Dense / Transit environments. This spatial pattern suggests that physical geographical barriers, such as coastlines, restrict alternative routes and are consistent with a stronger role of distance decay in commuting flows. We note, however, that these morphology-specific findings should be considered exploratory given the small sample sizes within each category.

To supplement this categorical classification, we analyze the continuous relationship between urban density and the performance gap ($\Delta\text{CPC} = \text{CPC}_{\text{DeepGravity}} - \text{CPC}_{\text{Proposed}}$). The gap correlates positively with both median employment density ($r = 0.263$) and population density ($r = 0.252$). This correlation indicates that dense, transit-oriented development decouples urban trips from simple Euclidean distance, allowing deep learning models to capture complex non-linear feature interactions, whereas lower-density or physically constrained environments follow more predictable distance friction behavior.

In summary, these results suggest that our aggregate-calibrated model provides a statistically equivalent alternative to locally-fit structures in low-density or physically bounded cities, offering a practical deployment pathway when local surveys are unavailable.

5 Discussion

The three research questions address the gaps identified in Table 1. The following discussion synthesizes the established findings and outlines practical implications of our decomposed gravity framework.

5.1 Empirical Evidence and Deconvolution Mechanics

Our results across 50 US metropolitan areas show that aggregate trip-length histograms preserve sufficient signal to estimate per-city distance decay curves. The recovered parameters correlate strongly with the power-law exponent γ ($r = 0.893$) and moderately with the exponential rate β ($r = 0.582$) fitted from complete OD flows. Under oracle outflows, our analysis indicates through a Two One-Sided Test (TOST) that the aggregate-calibrated gravity model is equivalent to the locally-calibrated Tanner model within a margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$). Additionally, Shapley analysis reveals that distance decay is the dominant transferable factor, accounting for 85.8% of the total CPC gain.

This recovery is explained by marginal projection under separability. Under separability, the OD matrix represents a joint distribution $P(X, Y)$ whose projection onto the distance domain D yields the TLD. Because the decay function $f(d)$ is the only component depending explicitly on zone distances, this projection preserves its shape. Although the spatial layout of population and POIs distorts the marginal distribution, the exposure-corrected MLE (Equation 5) deconvolves this geometric bias to isolate the latent decay parameters.

This deconvolution explains the comparative behavior under different settings. Under oracle outflows, the ground-truth-derived national prior slightly outperforms the binned Meta MDM calibration (CPC 0.7398 vs. 0.7337). Under GBDT-predicted outflows, this gap narrows to 0.0036 CPC (CPC 0.6896 vs. 0.6860), as the continuous national prior acts as a regularizer, smoothing out localized flow fluctuations and reducing sensitivity to production prediction noise. Critically, this comparison highlights a key conceptual contribution: the objective of localized aggregate recovery is not to outperform a rich national prior in data-abundant regions (such as the United States), but rather to serve as a direct replacement where such a prior does not exist. Constructing a national prior requires extensive historical commuting surveys across multiple representative source cities, which is a luxury unavailable in developing nations or data-scarce territories. In these zero-data deployment scenarios, the global availability of platform-sourced telemetry (e.g., Meta MDM) offers an essential, training-free calibration signal that enables zero-shot flow prediction where it was previously impossible.

5.2 Practical Deployment Implications

By demonstrating that high-fidelity decay calibration is achievable using only target-city aggregate histograms, this study enables a new paradigm for deploying mobility models. Calibrating per-city gravity models traditionally required invasive individual-level GPS telemetry or costly surveys. By combining globally available open-data features (OSM, WorldPop) with target-city aggregate TLDs, planning agencies can deploy structured gravity models in data-scarce or highly regulated environments. This pipeline bypasses the need for local travel surveys while preserving data confidentiality.

6 Threats to Validity

To ensure objective interpretation, we discuss the core limitations of our transfer learning workflow. First, *target mobility dependency* remains a concern. Under deployed GBDT outflows, the national prior fallback slightly outperforms localized calibration (CPC 0.6896 vs. 0.6860), indicating that localized calibration from binned telemetry does not necessarily yield downstream gains compared to a robust national prior. However, this concern is mitigated by the fact that a national prior cannot be constructed in data-sparse domains, leaving localized aggregate calibration as the only viable deployment pathway in such contexts. Second, *data source mismatches* and telemetry aggregation effects are the primary drivers of performance loss rather than bin compression. Decomposing the CPC drop shows that compressing LODES data from 20 bins to 3 bins only accounts for a minor decrease of 0.0055 CPC, while moving from survey-derived 3-bin data to real Meta telemetry results in a larger drop of 0.0268 CPC. Third, *sample and geographic constraints* limit generalization. The morphological classification contains small sample sizes (e.g., Coastal cities, $n = 3$), and commuting patterns in our US dataset may not generalize to European or Asian contexts with dense transit networks. Finally, *metric and geometric simplifications* affect accuracy. CPC is a global index blind to localized error structures, and using Euclidean distance ignores routing circuity and congestion travel times, which decouple travel from geometric distance in dense transit environments. Lastly, because our empirical evaluation is restricted to cities in the United States, commuting patterns, density profiles, and land-use arrangements may not generalize to European, Asian, or South American metropolitan contexts.

7 Conclusions

This study investigates the zero-shot cross-city transfer problem through a three-tier conceptual architecture, moving away from ad-hoc calibration toward a structural understanding of urban spatial interaction. Our findings across 50 US metropolitan areas indicate that the city-specific distance-decay function represents a dominant transferable factor of spatial flows. Furthermore, we show that the aggregate trip-length distribution (TLD) provides a privacy-preserving summary representation sufficient to identify this factor, enabling its successful application to zero-shot gravity model calibration.

7.1 Summary of Empirical Findings

We demonstrate that the target-city aggregate TLD—even when heavily compressed into a small number of physical distance bins—contains sufficient signal to recover the parameters of the distance-decay function. Under controlled oracle outflows, calibration on this aggregate measurement yields downstream prediction performance statistically equivalent to models calibrated directly on target-city individual-level OD flows. Furthermore, a game-theoretic Shapley analysis shows that distance decay dominates predictive capacity within this decomposed framework, reinforcing its role as the dominant transferable factor. Although localized calibration from real-world telemetry does not significantly outperform a robust national decay fallback, it does not require historical survey data from source cities, thereby enabling zero-shot deployment in data-scarce regions where such national priors cannot be constructed.

7.2 Theoretical and Practical Implications

These findings redefine how spatial interaction models can be calibrated. Historically, identifying city-specific travel friction required highly sensitive individual-level surveys or proprietary cellular trajectories. By demonstrating that the distance decay function can be robustly recovered using a one-dimensional aggregate representation (the TLD), we provide a deployable, survey-free forecasting pipeline. This architecture enables the calibration of structured, interpretable gravity models using only globally available open spatial data (OSM, WorldPop) and privacy-compliant aggregate indicators, facilitating deployment under privacy and data-governance constraints and reducing survey costs.

7.3 Limitations and Future Directions

Despite these contributions, several challenges remain. The separability approximation assumes independent spatial components, which may exhibit non-linear coupling in reality. Modeling this coupling while preserving zero-shot transferability is an open question. Second, the empirical evidence is restricted to US metropolitan areas characterized by high private vehicle usage; transferability to highly dense, multi-modal transit networks or informal settlements requires further cross-regional verification. Finally, using Euclidean distance ignores network routing circuity and congestion travel times. Incorporating these network and temporal dynamics into the aggregate calibration process represents an essential direction for future work.

Appendix: Supplementary Materials

Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

A.1 Estimation Details

To solve the maximum likelihood estimation (MLE) problem for target-city decay parameters $\theta = (\gamma, \beta)$ without disaggregate flows, we adopt a robust and physically grounded initialization and optimization workflow based on closed-form moment estimates.

Good parameter initialization is critical for optimization convergence. For the exponential decay rate β, we utilize a closed-form moment estimate derived under the assumption of pure exponential deterrence and a uniform distribution of spatial opportunities. In this theoretical limit, the expected travel distance satisfies $\mathbb{E}[d] = 1/\beta$. We estimate the empirical mean trip distance $\bar{d}_{\text{obs}}$ from the observed distance-bin histogram:

$$\bar{d}_{\text{obs}} = \sum_{k=1}^{K} b_k \cdot m_k \tag{9}$$

where b_k is the observed proportion of trips in bin k, and m_k is the midpoint of distance bin k (for the open-ended final bin, we use 1.5 times the lower boundary as a surrogate midpoint). The initial value $\beta^{(0)}$ is set as:

$$\beta^{(0)} = \frac{1}{\bar{d}_{\text{obs}}} \tag{10}$$

For the power-law exponent γ, the initial value is set statically to $\gamma^{(0)} = 1.0$, which corresponds to the center of the empirically observed range $[0.5, 2.0]$ in classical spatial interaction modeling literature.

The positivity constraints $\beta > 0$ and $\gamma \geq 0$ are strictly enforced during optimization via a parameter transformation. We define log-transformed parameters:

$$\tilde{\beta} = \log \beta, \quad \tilde{\gamma} = \log \gamma \tag{11}$$

and optimize the objective in the unconstrained log-transformed space $\theta_{\log} = (\tilde{\gamma}, \tilde{\beta}) \in \mathbb{R}^2$.

To guard against convergence to local optima due to the potential non-convexity of the exposure-corrected likelihood surface (especially under highly binned or noisy data), we run the L-BFGS-B optimization algorithm from 10 random restarts. The first run is initialized at the theoretical values $\theta_{\log}^{(0)} = (\log \gamma^{(0)}, \log \beta^{(0)})$. The remaining 9 runs are initialized by perturbing this central starting point with Gaussian noise:

$$\begin{aligned} \tilde{\gamma}_{\text{start}}^{(r)} &\sim \mathcal{N}(\log \gamma^{(0)},\, 0.25), \\ \tilde{\beta}_{\text{start}}^{(r)} &\sim \mathcal{N}(\log \beta^{(0)},\, 0.25) \end{aligned} \tag{12}$$

The optimization is executed for each restart run, and the parameter set yielding the highest likelihood value $\mathcal{L}(\theta)$ (Equation 5) is recovered via exponential back-transformation: $\hat{\gamma} = \exp(\tilde{\gamma}^*)$ and $\hat{\beta} = \exp(\tilde{\beta}^*)$.

A.2 Robustness Analysis

To evaluate the reliability of the exposure-corrected maximum likelihood estimation (MLE) workflow under non-ideal data collection conditions, we conduct a series of robustness analyses across four scenarios:

- **Bin Sensitivity Analysis**: Sweeping the number of distance bins $K \in \{3, 5, 10, 20\}$ reveals that downstream prediction performance stabilizes quickly. The mean CPC across target cities increases from 0.7174 under a coarse $K = 3$ configuration to 0.7346 for $K = 5$, 0.7394 for $K = 10$, and 0.7403 for $K = 20$ (compared to a baseline of 0.7411 computed on full, unbinned ground-truth OD matrices). This suggests that five or more distance bins provide sufficient resolution to capture city-specific decay profiles.
- **Atlanta Case Study Outlier**: Atlanta is identified as a localized outlier, where its Meta MDM 3-bin CPC under oracle outflows is 0.5761 compared to 0.7334 under the LODES 3-bin baseline. This drop is driven by the spatial interaction between the sprawling polycentric layout of the city and the coarse telemetry binning of the Meta Location Fraction data. Excluding Atlanta from the target set increases the mean target CPC for Meta MDM from 0.7080 to 0.7135, recovering 96.4% of the corresponding LODES baseline.

- **Missing Attractiveness Data (A_j)**: We simulate spatial feature scarcity by randomly removing 10%, 20%, and 50% of target-city destination attraction values A_j. The parameter recovery remains stable, yielding a mean absolute error (MAE) of $\gamma_{\text{MAE}} = 0.0395$ under 20% missing data, compared to the baseline MAE of 0.0394.
- **CBD Collapse (Extreme A_j Distortion)**: To test resilience against extreme structural distortions, we replace the attraction values A_j of the top 10% most attractive zones with the city-wide mean attraction. Under this CBD collapse scenario, the estimation remains robust, yielding parameter recovery errors of $\gamma_{\text{MAE}} = 0.0441$ under direct replacement and $\gamma_{\text{MAE}} = 0.0486$ when the affected zone attractions are scaled down by 80%.

A.3 Preprocessing and Normalization

Data preprocessing details are summarized in Table 9.

Appendix B. Robustness of Origin Production Transfer

We analyze the sensitivity of the zero-shot origin production transfer along two operational dimensions: training-set scale and prediction noise.

B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot outflow predictability by retraining the GBDT model on subsets of $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Evaluating on target cities demonstrates that zero-shot prediction performance rises steeply from $n_{\text{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Under substantial injected noise (40%), the target CPC degrades from 0.6896 to 0.6626. While this -3.9% drop is larger than the isolated production component contribution (3.6% Shapley value), it demonstrates that even highly corrupted outflow inputs do not collapse the spatial interaction matrix, because the production-constraint normalization restricts the error propagation.

Appendix C. Feature Selection Sensitivity

C.1 Feature Taxonomy and Candidate Set (37 dimensions)

We initially construct 37 candidate features per zone. This candidate space comprises 5 base scale and density variables (zone area, road density, residential population, total POI counts, and POI density) and 32 POI category count and density features (16 category counts and 16 category densities). The feature taxonomy is shown in Fig. 6.

C.2 SHAP-Based Feature Stability Selection

To identify features that generalize across cities, we apply SHAP importance analysis (Lundberg and Lee 2017) to the GBDT production model. Features are ranked by a cross-city stability score $S_p = \mu_p \,/\, (\sigma_p + \epsilon)$, where μ_p and σ_p denote the mean and standard deviation of per-city GBDT SHAP importances and $\epsilon = 10^{-5}$. A feature is retained if $\mu_p \geq 0.005$ and $S_p \geq 1.0$, ensuring both relevance and cross-city consistency. This analysis identifies 25 stable features from the 37-candidate space (32.4% dimensionality reduction) and confirms that the 6 core structural features used in the deployed model (Section 3.3) are among the primary and most stable predictors, with no accuracy loss relative to the full candidate set. The GBDT hyperparameter configurations are detailed in Table 10.

C.3 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 10, 15, 25, 37\}$) shows that predictive performance plateaus between 5 and 15 features (CPC: 0.7120). This indicates that most transferable spatial information is captured by a compact set

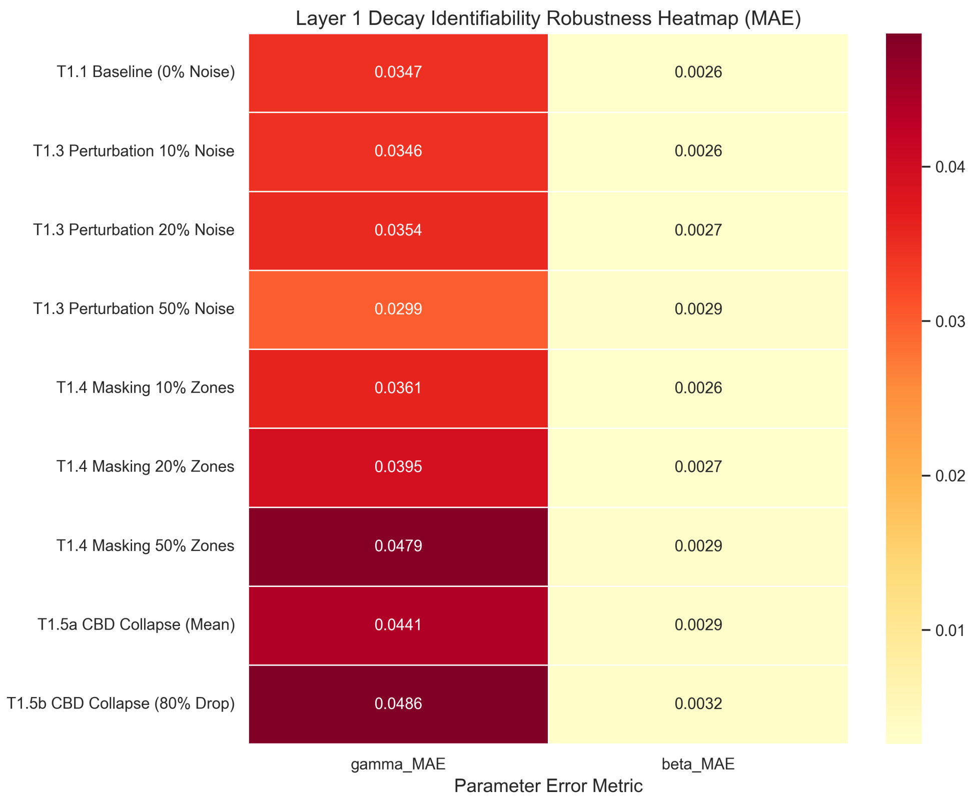

Fig. 4 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

Table 9 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	Treatment	Rationale
Missing Data	Median imputation	Rare ($< 2\%$); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

of built-environment indicators, with additional features providing marginal utility.

Appendix D. List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 11.

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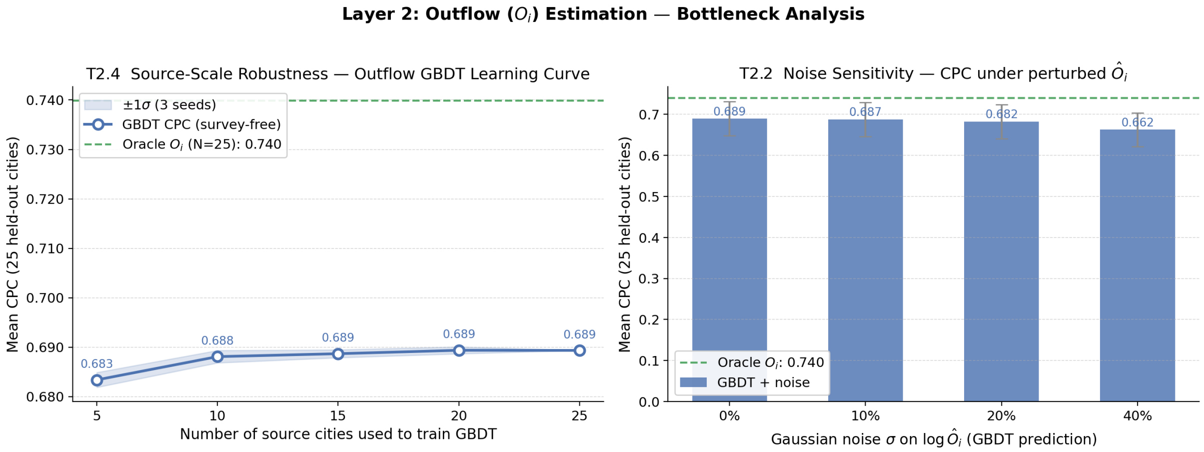

Fig. 5 Outflow bottleneck analysis. **Left**: CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

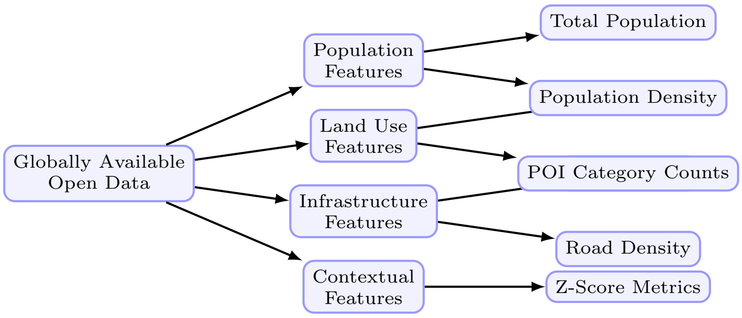

Fig. 6 Taxonomy of the globally available open data features used in the outflow model.

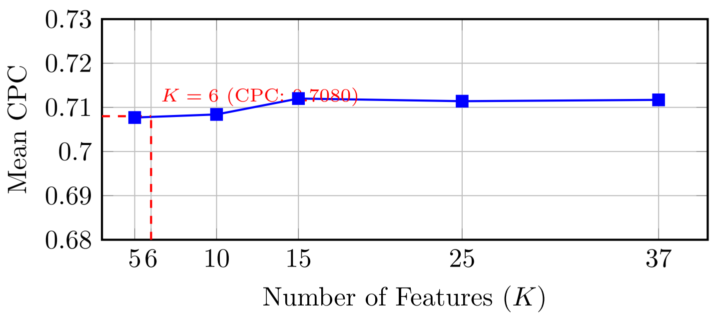

Fig. 7 SHAP feature count sweep over the 37-candidate feature set. Although utilizing more than 6 features yields marginally higher accuracy, it introduces a significant risk of overfitting to localized built-environment features.

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Table 10 GBDT Model Hyperparameter Configurations

Hyperparameter	**Outflow Model (O_i)**	**Attraction Model (A_j, ablation only)**
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	2	2
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

Table 11 List of Abbreviations

Abbreviation	**Definition**	**Abbreviation**	**Definition**
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations
SDF-ZTF	Separable Decomposed Formulation for Zero-Shot Transferable Flows		

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